Animating Digital Arithmetic

Will Dargan

November 11, 2026

Abstract

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Introduction

This project revolves around an application that enables the user to dynamically generate educational animations of simple arithmetic and bitwise logic on binary numbers.

Aims

Objectives

Chapter 1

Literature Review

1.1 Academic Theory

1.2 Technology Stack

Chapter 2

Design

2.1 Software Architecture

2.2 Requirements

2.2.1 Essential Features

ID	Name	Description
E-ANM01-F	Animating Addition	The system must generate animations of addition in binary.
E-ANM02-F	Animating Subtraction	The system must generate animations of subtraction in binary.
E-ANM03-F	Animating Multiplication	The system must generate animations of multiplication in binary.
E-ANM04-F	Animating Division	The system must generate animations of division in binary.
E-ANM06-F	Arithmetic Inputs	The system must allow the operands to be input by the user.
E-ANM05-F	Minimum Binary Precision	The system must allow for operands of at least 8-bits.
E-ANM07-N	Output Usability	The system must generate standalone video files as output.
E-GUI01-F	GUI Inclusion	The system must include a GUI.
E-ANM08-F	Written Subtraction	The system must generate animations of the long subtraction method with carries.
E-ANM09-F	Computed Subtraction	The system must generate animations of the two's compliment method of subtraction.
E-ANM10-F	Written Addition	The system must generation animations of the long addition method with carries.

2.2.2 Desirable Features

ID	Name	Description
D-GUI02-F	GUI Blocking	The generation of animations must not block execution of the GUI.
D-GUI03-F	Progress Bar	The system should display a progress bar when generating animations.
D-GUI04-F	Animation Preview	The system should offer the user the option to view a lower quality preview of the animation they are generating.

Chapter 3

Project Plan